

Some Topological Non-Realizability Results

Nick Kuhn — September 11, 2008.

Transcribed from three-page handwritten notes. [Download the printable transcript \(PDF\)](#). Diagram arrows and annotations are described in words as well as formulas.

Page 1: Realizing graded rings

Given a space X , its cohomology $H^*(X)$ is a graded-commutative ring. That is, $H^*(-)$ is a functor from the homotopy category of spaces to graded-commutative rings, written in the notes as

$$\mathrm{HoTop} \rightarrow \mathrm{GrComRing}.$$

Question. What graded rings can be topologically realized? Given R^* , is there a space X with $H^*(X) = R^*$?

The most famous version is the **Steenrod problem (1961)**: if

$$H^*(X) = \mathbb{Z}[y_{2d_1}, \dots, y_{2d_n}],$$

what can $d_1, \dots, d_n$ be?

Examples.

$$H^*(\mathbb{CP}^\infty) = \mathbb{Z}[x_2],$$

$$H^*(BSU(n)) = \mathbb{Z}[x_4, x_6, \dots, x_{2n}], \quad x_{2i} = c_i,$$

$$H^*(BSp(n)) = \mathbb{Z}[x_4, x_8, \dots, x_{4n}], \quad x_{4i} = p_i.$$

Theorem (Andersen–Grodal, 2007). If $H^*(X)$ is polynomial, it is isomorphic to a product of these. This is closely related to the theory of finite loop spaces.

One can pose these questions with coefficients in a field $\mathbb{F}$. In characteristic zero,

$$H^*(\Omega S^{2n+1}; \mathbb{Q}) = \mathbb{Q}[x_{2n}],$$

so any polynomial algebra over $\mathbb{Q}$ works.

Characteristic $p > 0$ is much more exciting: mod p cohomology has extra structure as a module over $\mathcal{A}$, the Steenrod algebra.

For $p = 2$, there are natural group homomorphisms

$$\mathrm{Sq}^k : H^*(-) \rightarrow H^{*+k}(-).$$

The algebra $\mathcal{A}$ is generated by Sq^k , for $0 \leq k < \infty$, subject to the Adem relations. It has the same size as a polynomial algebra with generators in degrees $2^k - 1$, for $k \geq 1$.

Then $H^*(X; \mathbb{F}_2)$ is an $\mathcal{A}$ -module.

Page 2: Unstable modules and unstable algebras

The module satisfies the unstable condition: if $x \in H^n(X; \mathbb{F}_2)$, then

$$\mathrm{Sq}^i x = 0 \quad \text{for } i > n.$$

Thus $H^*(X)$ belongs to $\mathcal{U}$, the category of unstable $\mathcal{A}$ -modules. The diagram shows $H^*(-)$ taking values both in $\mathcal{U}$ and in graded-commutative algebras; these structures interact.

Question 2. What unstable $\mathcal{A}$ -modules can be realized?

1. Cartan formula:

$$\mathrm{Sq}^k(x \cup y) = \sum_{i=0}^k \mathrm{Sq}^i(x) \cup \mathrm{Sq}^{k-i}(y).$$

[Transcription note: the scan writes x in the second factor on the right; that argument is y in the Cartan formula.]

2. Restriction axiom: if $x \in H^k(X)$, then

$$\mathrm{Sq}^k(x) = x \cup x.$$

A margin annotation compares this with a restricted Lie algebra. Steenrod called these the “reduced squares.”

Example.

$$H^*(\mathbb{RP}^\infty; \mathbb{F}_2) = \mathbb{F}_2[t], \quad t \in H^1,$$

$$\mathrm{Sq}^k(t^n) = \binom{n}{k} t^{n+k}.$$

An **unstable algebra** is an algebra which is an unstable $\mathcal{A}$ -module satisfying the Cartan formula and the restriction axiom. Let $\mathcal{K}$ be the category of unstable $\mathcal{A}$ -algebras. Then

$$H^*(-; \mathbb{F}_2) : \mathrm{HoTop} \rightarrow \mathcal{K}.$$

The $\mathcal{A}$ -module structure restricts the algebra structure.

Example. There does not exist X such that $H^*(X) = \mathbb{F}_2[x_3]$. Indeed,

$$\mathrm{Sq}^3 x_3 = x_3^2 \neq 0,$$

but

$$\mathrm{Sq}^3 x_3 = \mathrm{Sq}^1 \mathrm{Sq}^2 x_3 = \mathrm{Sq}^1(0)$$

for degree reasons.

The algebra structure also restricts the module structure.

Example. Consider $\mathcal{A} \cdot t \subseteq \mathbb{F}_2[t]$, denoted

$$\phi = \langle t, t^2, t^4, t^8, \dots \rangle.$$

The diagram joins consecutive generators by Sq^1 , Sq^2 , Sq^4 , and so on.

Is ϕ the cohomology of a space? No. Writing its classes as $x_1, x_2, x_4, \dots$, the relations

$$\mathrm{Sq}^1 x_1 = x_2, \quad \mathrm{Sq}^2 x_2 = x_4$$

would imply

$$x_2 = x_1^2, \quad x_4 = x_2^2 = x_1^4,$$

but there is no class x_1^3 .

Page 3: Suspension and non-realizability

Conjecture. If M is a finitely generated unstable $\mathcal{A}$ -module which is infinite-dimensional over $\mathbb{F}_2$, there does not exist X with

$$\tilde{H}^*(X) \cong M.$$

Theorem (Kuhn–Schwartz). This is true.

The suspension ΣX is drawn as two cones on X glued along their common base. There is a suspension endofunctor of $\mathcal{U}$, defined by

$$(\Sigma M)^n = M^{n-1}.$$

The notes record

$$H^*(\Sigma X) = \Sigma H^*(X),$$

and that all cup products vanish.

Does there exist a space with $H^*(X) = \Sigma \phi$? No.

1. A hard proof uses the chain of operations Sq^1, Sq^2, Sq^4 , and so on. The diagram singles out an arrow labeled Sq^{16} . A margin annotation says an additional cell would be needed there by the Hopf invariant one result.
2. Lionel shows that a three-stage part cannot exist. The diagram for a putative space Z has classes x_{17} , x_{33} , and x_{65} , with successive arrows from the first to the second and from the second to the third. Try to compute $H^*(\Omega Z)$; enough can be obtained to get a contradiction.

Then replace $\mathbb{Z}/2$ by M . The next diagram has three copies of M , with arrows labeled Sq^{2^k} and $Sq^{2^{k+1}}$.

One needs many loops, depending on $\dim M$ and other data. A recent paper allows an easier argument using the map written in the notes as

$$Z \rightarrow \Sigma^\infty \Omega^n Z.$$